

SECTION 16276-A
OIL - IMMERSSED POWER TRANSFORMERS

PART 1 - GENERAL

1.1 DESCRIPTION

This specification covers designing, furnishing, installing, and testing of two sets of 25/31.25 MVA, OA/FA, 22.8KV/6.6KV oil immersed power transformer.

1.2 APPLICABLE CODES AND STANDARDS

The codes and standards applicable to only a portion of the work specified in this section are referenced in the relevant paragraphs. Codes and standards which are generally applicable to the work of this section are listed below.

Sponsor	Number	Title
IEC	60076-1	Power Transformer, Part 1: General
IEC	60076-2	Power Transformer, Part 2: Temperature Rise
IEC	60076-3	Power Transformer, Part 3: Insulation Levels and Dielectric Tests
IEC	60076-4	Power Transformer, Part 4: Tappings and Connections
IEC	60076-5	Power Transformer, Part 5: Ability to Withstand Short Circuit
ANSI	C57.12.00	General Requirements for Liquid Immersed Distribution, Power, and Regulating Transformers
ANSI	C57.12.10	Requirements for Transformers 230,000 Volts and Below; 833/958 Through 8,333/10,417 KVA, Single-Phase, and 750/862 Through 60,000/

80,000/100,000 KVA, Three Phase

ANSI	C57.12.90	Test Code for Liquid -Immersed Distribution, Power, and Regulating Transformers and Guide for Short-Circuit Testing of Distribution and Power Transformers
ANSI	C57.92	Guide for Loading Mineral - oil Immersed Power Transformers Up to 100 MVA with 55°C or 65°C winding Rise
ANSI	C62.1	IEEE Standard for Surge Arresters for AC Power Circuits
ASTM	D117	Electrical Insulating Oils of Petroleum Origin, Method of Testing
CNS	598	Power Transformer
CNS	599	Testing Standard of Distribution Transformer
IEEE	21	IEEE General Requirements and Test Procedure for Outdoor Apparatus Bushings
NEMA	CC1	Electrical Power Connectors for Substations
NEMA	TR1	Transformers, Regulators and Reactors
NEMA	TR98	Guide for Loading Oil-Immersed Power Transformers with 65°C average winding Rise
NEMA	107	Method of Measurement of Radio Influence Voltage (RIV) of High-Voltage Apparatus
NFPA	70	National Electrical Code

Where references are made to standards, the Contractor may submit equipment built and tested according to other standards providing that the requirements of the referenced standards are complied with as a minimum.

1.3 SUBMITTALS

A. Bidder Submittals

1. Data: Power transformer specification sheet as shown in appendix of ANSI C57.12.00.
2. Certification: Certification that the transformers will meet the short circuit withstand requirements of ANSI C57.12.00 (1987 revision), section 7.
3. Physical dimensions, weight and detail drawing showing all accessories.

B. Contractor Submittals

1. Certification that oil has been tested in accordance with ASTM D117.
2. Certification that transformers has been tested as indicated in section 2.3 - shop tests.
3. Submittal Schedule.
4. Installation schedule: Including fabrication, transportation, erection and testing.
5. Seismic-resistance calculation.
6. Shop and field test procedures.
7. Test Reports.
8. Calculation of Annual Maintenance Cost.
9. Maintenance Schedule.

1.4 SHIPPING, HANDLING AND STORAGE

A. Storage

1. Measure and record the oil level and the ambient temperature when the unit is received and every month thereafter. If the level falls, search for leaks and add insulating liquid to keep the level within tolerances.

1.5 WORKS

A. Works Included

1. Furnish two sets of transformers. Ratings as indicated in this specification.
2. Install transformers on the foundation.
3. Test transformers.

B. Works Excluded

1. Concrete foundation
2. Protective relays except these specified in this specification.
3. Online partial discharger monitoring
 - a. To use ultra high frequency (UHF) method.
 - b. Online partial charger sensor and connect to PD system.

1.6 GUARANTEE

If any part of the material or equipments is found defective due to design, workmanship, or quality of material within two(2) years after shipment, the contractor shall replace the defective parts within one(1) week, including workmanship and customer's duty if levied, at the contractor's expense. The guarantee period should expand two(2) years (from Oil-Immersed Power Transformers recover day).

PART 2 - PRODUCTS

2.1 SERVICE CONDITIONS

The transformers will be used under the following conditions:

A. Altitude

A height above sea-level not exceeding 1000m.

B. Temperature of Cooling Medium

Air at a temperature never exceeding 40°C, and never below 0°C; and the air temperature never exceeding the following values:

33°C average in any one day;

20°C average in any one year.

C. Wave Shape of Supply Voltage

A supply voltage of which the wave shape is approximately sinusoidal.

D. Symmetry of Polyphase Supply Voltages

For polyphase transformers, supply voltages which are approximately symmetrical.

2.2 DESIGN REQUIREMENTS

A. General: Transformers shall be designed, constructed, rated, and shall perform in accordance with ANSI C57.12.10. Transformer shall be able to withstand Short circuit current according to the latest ANSI standard C57.12.00. The short-circuit current rating at the 22.8kV winding side is designed at 25kA and the short circuit withstand time shall be not less than 2 sec. for three-phase faults occurred at 6.6kV side.

B. Transformer Characteristics: Power transformers, two winding, 3 phase, 60 Hz, with the following characteristics:

1. Insulation Mineral oil immersed

2. Output, MVA 25/31.25 MVA, 75°C OA/FA

3. Winding temperature rise above 40°C ambient measured by resistance method shall be less than 65°C.

4. Oil temperature rise above 40°C ambient measured by thermometer method shall be less than 65°C.

5. Rated voltage/connection:
 - a. HV 22.8 KV delta

 - b. LV 6.6 KV, wye, low resistance grounded

6. Insulation class:
 - a. Winding BIL
 - (1) HV

Crest, 1.2x50 μ s	150 KV
full wave	

 - (2) LV

Crest, 1.2x50 μ s	75 KV
full wave	

 - b. Winding, low frequency
 - (1) HV

Low frequency dielectric test voltage	50 KV
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 - (2) LV & Neutral

Low frequency dielectric test voltage	26 KV
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 - c. Bushing BIL
 - (1) HV

Crest, 1.2x50 μ s	150 KV
full wave	

 - (2) LV

Crest, $1.2 \times 50 \mu s$
full wave

75 KV

d. Bushing, low frequency

(1) HV

Low frequency dielectric test voltage (Liquid test 10 sec) 50 KV

Low frequency dielectric test voltage (Dry test 1 min) 60 KV

(2) LV

Low frequency dielectric test voltage (Liquid test 10 sec) 24 KV

Low frequency dielectric test voltage (Dry test 1 min) 27 KV

7. Tap changer:

a. Type No-load tap changer

b. Positions Four positions minimum
(2 positions are above normal and 2 positions are below normal, +5.0%, +2.5%, 0, -2.5%, -5%)

8. Percent impedance 9.0% minimum at OA rating.

9. Polarity Negative

C. Temperature Rise and Loading

1. The transformer shall be designed to permit loading in accordance with ANSI C57.92.

2. Loading shall not be limited by accessory equipment such as tap-changers, or terminal pads.

3. Transformers shall provide rated load at a daily average ambient temperature of 33°C and a maximum ambient temperature of 40°C , without exceeding average

winding temperature rise of 65°C and top oil temperature rise of 65°C.

4. The transformer shall be capable of supplying rated load without overheating when the load is made-up of 100% normal 60Hz sine-wave current plus the following harmonic currents:
 - a. 16% of the fundamental as 3rd harmonic current
 - b. 10% of the fundamental as the 5th harmonic current
 - c. 7% of the fundamental as the 7th harmonic current
 - d. 5.5% of the fundamental as 11th harmonic current
 - e. Smaller percentages through the 25th harmonic current

D. Taps

1. The transformer shall be furnished with full capacity high-voltage taps. The tap changer shall be clearly labeled to reflect that the transformer shall be de-energized before operating the tap changer as required in Section 5.1.1 of ANSI C57.12.10. The tap changer switch shall be an eternally operated, samp action switch with a level handle. Padbcking provisions are standard.
2. The unit shall have the tap configurations of two-2½% taps above and below rated voltage.

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E. Tank, Base and Control Cabinet

1. The transformer tank shall be of welded steel construction and outside painting shall be of corrosion resistant.
2. The centerlines of tank or case shall be permanently marked near the base on all four sides of the tank or case and shall also be shown on the outline drawing.
3. The gasketed surfaces around openings in the transformer cover for mounting of items such as bushings, manholes and relief devices, shall be raised sufficiently to prevent accumulation of water around the gasket.
4. All enclosures shall be weatherproof and designed to prevent accumulation of standing water along the panel doors. Gaskets of all removable cover-plates and doors shall be recessed and provided with

mechanical gasket stops or shielding for adequate protection from the weather.

5. The control cabinet shall have one door and shall be located under the conservator within reach of a person standing at the transformer base level. The control cabinet shall contain all auxiliary devices except those that must be located directly on the transformer. Doors shall be hinged and provided with stops to hold the doors in the open position until released by the operator. Auxiliary panels shall be hinged or otherwise designed for ready accessibility to relays, control devices, and associated terminals for testing, adjustment, and maintenance.
6. As a minimum, alarm devices shall be as follows, the alarm conditions shall be displayed on local panel.
 - a. Cooling device control power under voltage indication 1 circuit°
 - b. 480 Volt main power trip indication 1 circuit°
 - c. (Buchholz Relay) Alarm indication 1 circuit°
 - d. (Buchholz Relay) Trip indication 1 circuit°
 - e. (Pressure Relief Device) indication 1 circuit°
 - f. (Oil Sudden Pressure Relay) trip indication 1 circuit°
 - g. High oil temperature alarm indication 1 circuit°
 - h. High voltage side winding high temperature alarm 1 circuit°
 - i. Low voltage side winding high temperature alarm 1 circuit°
 - j. Low Oil surface indicator 1 circuit°
 - k. DC control power supply trip indication 1 circuit°
 - l. spare : 5 circuits°

F. Core and Coils

1. HV windings shall be delta connected.
2. LV windings shall be wye connected.
3. The transformer core shall be grounded to the transformer tank. To facilitate testing of core clamp insulation, the core ground shall be accessible from a manhole. The ground strap shall be slotted for disconnection without complete removal of hardware and without lowering the oil level.

4. The transformers shall be provided with a thermally upgraded insulation system which permits continuous operation without loss of insulation life at the specified ratings.
5. Windings must be surge-proof with the best protection against electrical surges.

G. Terminations

1. The high voltage terminations shall be porcelain outdoor type, oil insulated, impulse tested bushing with bolted type connectors.
2. The terminations voltage shall be wall mounted in a air insulated junction box. Junction box shall be provided for the XLPE single core cable entrance.
3. The position of bushings shall be verified with the Owner prior to manufacturing.

H. Oil

1. Oil shall be insulating mineral oil to the IEC 296 Class 1.
2. Transformers shall be filled to the 25°C liquid level mark.
3. Any transformer oil supplied for topping-up purposes shall be shipped in sealed drums which shall remain sealed until required.
4. Manufacturers should provide enough transformer insulation oil required for 110% of the total amount without oxygen, and shall be pure hydrocarbon mineral oil.

I. Oil Conservator

1. Conservator shall be provided with cleaning doors, oil level gauge and valves. The outlet pipe shall protrude at least 25mm into the conservator so as to trap any sediment or moisture within the conservator and not allow these to enter the main tank.

2. The conservator tank shall be fitted with a breather in which silica gel is the dehydrating agent. The breather shall be a moulded type transparent case of adequate size.
3. Conservators shall be mounted at the end of the transformers in such a way as not to construct external connections or reduce the safety clearances to live external connections.

J. Cooling Systems

1. The transformers shall be the manufacturer's standard design suitable for the duty specified, oil immersed, self-cooled (OA), with one stage of force air cooling (FA). Fans and controls shall comply with ANSI C57.12.10 section 12.3.8 for automatic control by winding temperature except that fan motors shall be three phase if 0.37 KW or larger. The control system for the cooling equipment shall be enclosed in a NEMA 3 self standing cabinet installed near the transformer. Cooling fans need to be able to operate manually / automatically and can switch between operation. The fan operating control voltage will be 3 phase, 4 wire, 480v. The feeder will be brought to the main lugs of the control panels by others.

K. Wiring

1. Control, auxiliary power, and alarm circuits shall be completely wired. Wiring shall be run in rigid metal raceways, except that motor connections shall include flexible interlocked, armored, liquid-tight conduit.
2. Enclosures for mounting the terminal blocks shall be provided and mounted near the base of the transformer. Conduit connections shall clear the box for removal of transformer.
3. Terminations shall be segregated and separated by low-voltage power, low-voltage control, and metering.
4. Control and alarm switches shall not be located more than 2m nor less than 0.75m from the transformer base.

L. Oil deterioration preventing device :

The transformer should use air bag type oil deterioration prevention device to completely isolate the oil surface from the air. Deterioration prevention

devices must be accompanied by maintenance-free respirators and other necessary accessories.

M. Gas detection device :

The transformer shall be provided with a gas detection device that communicates with the Buchholz Relay via a connecting pipe and valve so that the gas can be collected for inspection and analysis while the transformer is running.

The gas detection device should be installed in the proper position and height (about 1.0~1.3 m from the ground) to make it easy for the staff to sample on the ground.

The gas detection device shall be provided with a transparent glass inspection window for the inspection of the faulty gas. A gas collection valve and a drain valve shall be provided at the top and bottom of the inspection window respectively.

N. Accessories

1. A bottom drain valve and a built-in liquid sampling device shall be provided. This valve shall permit complete drainage of the tank.
2. A top filter press connection valve shall be provided.
3. A pipe and cap on top of the main tank, but remote from the top filter press, for vacuum connection during vacuum filling shall be provided.
4. Cabinets containing indication instruments, such as pressure gauges, shall have windows to view these instruments.
5. A plugged or capped pipe fitting shall be provided at both the top and bottom of radiators and coolers. Plugs shall be solid bronze or cast iron with a square head.
6. Buchholtz relay - 1st & 2nd stage relay shall be provided.
7. A pt 100 at 0°C resistance temperature detector in the low-voltage windings shall be provided.
8. Two copper faced grounding pads shall be provided, located on opposite sides of the base, drilled and tapped for two tapped holes per NEMA CC1, Section 4.05 for attaching ground cable connector to the tank.

9. One clamp - type terminal for each grounding pad, for 150 to 300 mm² cable, shall be provided.
10. A self - resetting, mechanical trip, pressure - relief device with visual trip target and normally closed electrical alarm contacts suitable for 125V dc shall be provided.
11. Temperature and liquid-level indicators shall be fitted with contacts in accordance with ANSI C57.12.10, 12.3.6. Indicators shall be at eye level.
12. A top oil temperature indicator, dial thermometer type with minimum indicator and an external reset normally closed high temperature alarm contact for 125V dc, shall be provided. The thermometer shall have a scale range calibrated to 150°C.
13. A stainless steel diagram nameplate shall be provided in accordance with ANSI C57.12.00. Units shall be the SI system.
14. Thermostatically controlled space heaters shall be provided in a readily accessible location in all control cabinets and terminal chambers of the transformers. Heaters in locations that may be hazardous to operating personnel shall be provided with protective screens. Each space heater circuit shall be provided with a manual and a thermostatically controlled switch and shall be separately fused and coordinated to maintain selectivity with the main protective device.
15. All gaskets in contact with oil shall be nitrile rubber, or approved equivalent. Cork or corprene gaskets will not be acceptable.
16. All control cabinets shall be provided with internal illumination and a 120V ac, 20A outlet.
17. The dial type winding temperature relays with a range up to 150°C shall be provided. The different relays shall be used to operate cooling fans and to alarm.

O. Interface with SCADA System

The following alarm / status shall be extended to a terminal strip to allow for remote monitoring.

1. Transformer oil low level, dry contacts output;
2. Top oil high temperature, dry contacts output;
3. High voltage side and low voltage side Winding hot-spot high temperature, dry contacts output;
4. Pressure relief device trip, dry contacts output;
5. Buchholtz relay trip, dry contact output;
6. Top oil temperature, 4-20mA_{dc} output;
7. Winding hot-spot temperature, 4-20mA_{dc} output;
8. Cooling device control power under voltage, dry contacts output °
9. 480 Volt main power trip, dry contacts output °
10. (Buchholz Relay)Alarm, dry contacts output °
11. (Oil Sudden Pressure Relay)trip, dry contacts output °
12. High oil temperature alarm, dry contacts output °
13. Low Oil surface, dry contacts output °
14. DC control power supply trip, dry contacts output °
15. spare : 5 dry contacts output °

P. Special Requirements

1. The transformers will be bolted to the floor at ground level. They shall withstand seismic forces in accordance with China Building Technical Regulation for continued operation.
2. Average audible sound levels measured in accordance with ANSI C57.12.90 shall not exceed 75 dbA.
3. Gauges and indicators shall be mounted to be visible at ground level. Any device requiring adjustment shall be accessible from ground level or provisions for access provided.

4. If necessary oil ducts shall be included in the core design to maintain the core operating temperature within safe limits

2.3 SHOP TESTS

A. General

1. Transformer tests shall be performed in accordance with ANSI C57.12.90 and shall include all routine tests and specified other tests. Tests shall be performed with the specified terminations in place.
2. Items of shop test and their application refer to Appendix 1.
3. The contractor shall invite the owner or his representative to witness the shop tests. UIS or Micron needs to attend the FAT.

B. Full Load Temperature Rise Tests

1. Full load temperature rise tests shall be made on all transformer. Data listed below shall be obtained with the transformer connections giving the maximum heating:
 - a. Top oil temperature rise at rated voltage and rated load;
 - b. Average temperature rise by resistance at rated voltage and rated load for all windings;
 - c. Calculated hottest - spot temperature rise over average winding rise for each winding reported above.
2. Temperature tests shall be made with OA and FA cooling.
3. The ambient temperature shall be the idle room air temperature measured at least 3m from the transformer.
4. If test data for a transformer determined to be a duplicate are made available, temperature rise tests will not be required.

PART 3 - EXECUTION

3.1 FIELD TESTS

- A. Measurement of winding resistance at principal tap and two extreme taps.
- B. Voltage-ratio measurement and check of vector group.
- C. Power factor measurement of transformer coils, bushings, etc.
- D. Insulating oil characteristics test.
- E. Action test of various protection devices.
- F. Control loop function test.
- G. Test of polarity, phase angle and burden of the current transformer.

3.2 MANUFACTURER'S REPRESENTATIVE

The services of a manufacturer's representative shall be provided.

APPENDIX 1

SHOP TEST

TEST	APPLICATION
1. Appearance structure check	All transformers
2. Resistance measurements	All transformers
3. Ratio	All transformers
4. Polarity and phase relation	All transformers
5. No-load losses and excitation current, 100% rated voltage	All transformers
6. No-load losses and excitation current, 110% rated voltage	One transformer
7. Impedance voltage and load loss	All transformers
8. Current transformer test	
9. Partial discharge test	All transformers
10. Pressure relief device test	All transformers
11. Insulating oil test Transformer insulating oil dissolved gas test (including furfural) and characteristics test, 72HR after power energized and one month after the paralleling operation, the test vendor will be required with the national laboratory certification for transformer insulation oil test. (site test).	All transformers
12. Temperature rise	All transformers, both OA and FA cooling
13. Dielectric test	
(a) Low frequency, applied and induced voltage	All transformers
(b) Lightning impulse	All transformers
(c) Radio frequency voltage	All transformers

(d) Insulation resistance	All transformers
(e) Audio sound	All transformers
14. Leak	All transformers
15. Transformer winding deformation test (before delivery).	All transformers

TYPE TEST

1. Impulse voltage test (Impulse Test) Contains full wave and chopped wave two tests.
2. Temperature rise test

Note: *Test certificates shall include not less than the minimum data included in Appendix to Part 1 of ANSI C57.12.90 - 1987.

- END -